E-Cigarette and Healthy Habits Study

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Background and Question

Question: Are those with healthier lifestyles less likely to use e-cigarettes than those who may have less healthy lifestyles?

Hypothesis: We hypothesize a predictive model for e-cigarette usage can be built using health and lifestyle data because unhealthy habits frequently co-occur.

Prediction: We predict that those with healthier living habits may be less likely to use e-cigarettes due to their desire to live the healthiest lifestyle.

Methods

The dataset used to perform this analysis came from the 2024 Behavioral Risk Factor Surveillance System (BRFSS) from the Centers for Disease Control and Prevention (CDC) website. This dataset contains extensive health information from residents of the United States and its territories. Some of these include variables such as e-cigarette use, drug, tobacco, and alcohol use, demographics, physical and mental health, and lifestyle behaviors. The target variable for this analysis is E _CIG_USER, which represents whether the individual is a current e-cigarette user or not.

The data was originally in the form of a SAS Transport File (.XPT), which was subsequently read into R and transformed and outputted as a comma separated value (.CSV) file. This file was then read into Python, where the data was wrangled using the NUMPY and PANDAS packages. The data wrangling process consisted of making variable names easier to decipher, transforming observations from each variable from numerical values with no true meaning to numerical and categorical values with meaning, performing basic calculations to make variables more useful, and removing unnecessary variables. Once this was completed, the

dataset was outputted as a comma separated value (.CSV) to be used for the Exploratory Data Analysis.

Methods

The variables were split into categorical and continuous types to create the appropriate statistical summaries. The focus of the categorical variable analysis included the counts and percentages of each variable. The continuous variables were measured with multiple statistical methods including count, mean, standard deviation, median confidence interval, minimum, maximum, skew, and kurtosis. Additionally, tables summarizing the counts and percentages for missing or unknown records were created. These tables will allow the team to better understand which variables are missing extensive data. For example, our newly created variables Sugar_Soda_Drinks_Per_Day, Sugar_Soda_Drinks_Per_Week, Sugar_Soda_Drinks_Per_Month, Non_Soda_Sugar_Drinks_Per_Day, Non_Soda_Sugar_Drinks_Per_Week, and Non_Soda_Sugar_Drinks_Per_Month all have between 86%-97% missingness. They also have negative values in the statistical output for the mean, median, minimum, skew, and kurtosis. Both of these findings indicate that further analysis on these variables will be needed to determine their quality and value for answering our question. Additional continuous variables with high missingness include Average_Number_of_Drinks_30_Days (54.62%) and Marijuana_Use_30_Days (70.31%).

Next, the continuous variables were summarized by outcome classes. This table can be found in the appendix and allows us to look at the distributions, relationships, and statistics of these variables among their outcome classes. Notable findings from this table include the blank outcome class for the Number_of_Children variable having a kurtosis of 1,170.47 and the 0.0 outcome class for the Sugar_Soda_Drinks_Per_Day variable having a kurtosis of 174.15. They

also both have high skew: 22.78 and 12.57. These findings indicate that these fields are right-skewed and may have a larger number of outliers than other fields.

Lastly, boxplots and barplots were created for EDA. Boxplots were created to analyze continuous variables and their statistics. Barplots were created to analyze categorical variables, comparing proportions among different variables, such as age categories and sex. These tables were created so we can efficiently obtain helpful statistics and frequencies of the current variables. When combined, these tables and graphs help us understand our data better, find remaining data quality issues, and determine which variables will be essential in helping us answer our question.

Discussion & Next Steps

One of the key takeaways from the analysis thus far is that many of our continuous variables are highly skewed. Number_of_Children and Sugar_Soda_Drinks_Per_Day have the highest skew, whereas variables such as BMI have a much smaller skew. Since many of the variables are skewed, the median is an appropriate statistical measure. Another notable finding is that variables such as Number_of_Children, Marijuana_Use_30_Days, Poor_Physical_Health_Last_Month, and Poor_Mental_Health_Last_Month all have a median of zero.

Table 4 highlights that many of the continuous variables have missing values. Table 1 also shows that there are multiple categorical variables with missing values, such as Health_Insurance and E_Cig_User. Since E_Cig_User is our response variable, one of our next steps is to carefully decide how we will handle the missing records in this field.

Table 2 highlights unknown counts for categorical variables, including Loneliness, Prediabetic_Diagnosis, Social_Emotional_Support, and Life_Satisfaction. All of these variables have over 50% of the field missing. For both categorical and continuous variables, the missing and unknown values will need to be handled later in the analysis.

As seen in Figure 1, our preliminary EDA shows that there is a potential connection between e-cigarette usage and poor mental health days. This will need to be explored further, with additional mental health variables, to help answer our question of whether individuals with healthier lifestyles are less likely to use e-cigarettes. Additional analysis including demographics, physical health, and lifestyle variables will be performed to further strengthen our findings.

For the most part, our analysis plan remains the same. Now that we have cleaned our dataset and created some exploratory data analysis graphs and tables, we have noticed some inconsistencies with the data that we would like to fix as part of our Data Cleaning and Pre-Processing Plan. For example, there are some variables that were removed that we want to add back in. Some of the variables themselves require some adjustments in order to get them to more logical values. The BMI variable, for example, needs to be divided by 100 in order to get to a more reasonable value. Some of the variables will also need to be evaluated for their quality to see if they may remain valuable enough to keep. Aside from these changes, our analysis plan remains the same (attached below). We will then proceed with variable selection and creating the initial models for our analysis.

Analysis Plan

Initially we will be reducing the huge number of variables from the BRFSS dataset, with a focus on demographics, e-cigarette usage and a list of health based habits such as Heavy Alcohol Usage, Overweight or Obese and Exercise in the past 30 Days.

Then we will begin analyzing any missing data points and determining whether deleting these data points, filling them in with means or medians, or removing the variable entirely is appropriate. With such a large dataset to work with, we will have a number of options here without risking diluting our dataset down too low where the results will be skewed. Our initial plan is to change any blank, "refused", or "don't know" data points to NaN's, and any data listed as "nones" to zero.

Then we will analyze correlation between the variables. This step is crucial because it can lead you down different paths. Some sets of data can be so heavily correlated, regardless of variable selection, that it can determine your model selection. Other data sets will only have a handful of variables that are strongly correlated, which can lead to stricter variable selection, but a wider possibility of model options. For the purposes of this study, we will need to look at correlations between variables that measure very similar health measures in similar ways. For example, the measure exercise in the last 30 days and exercise used to assist with joint or arthritis pain. It's likely this process will lead to us only picking one of these variables.

Then we will be exploring the distribution of data in our remaining variables, to see if these are distributed in a bell curve, or have a spread of outliers. We will also discuss the value of changing certain variables into categories, such as turning BMI into underweight, healthy weight and overweight, to increase interpretability.

In regards to model selection, we are focused on a binary outcome, so we will first be using logistic regression, as this gives us easily interpretable impacts each variable had in predicting likelihood of e-cigarette usage.

As mentioned before, we also risk having too many variables, and ones that are heavily correlated, so this is motivating us to try a Ridge model, which will reduce the impact of

correlated variables and offer us stability in coefficients. Lastly, a Random Forest model can handle non-linear relationships and would offer a great comparison to our logistic regression model. For an unsupervised model, we plan to use K-Means-Clustering. This will allow us to see natural forming groups we would not otherwise notice in a supervised model.

In terms of determining if our hypothesis is correct, that healthier people are less likely to smoke e-cigarettes, we can evaluate this using an ROC curve to evaluate the predictive strength with the goal being the curve being as close as possible to the top left and an area under the curve in our plot above 80%.

Then we can produce a confusion matrix to analyze our performance in three key metrics: precision, recall and F1 score. Precision will be used to make sure we are not over predicting false positives, recall will be used to ensure we are minimizing false negatives, and F1 is used to make sure we are balancing each.

Due to different strengths within our team, we plan to use a combination of R and Python for this project.

Results

When analyzing potential variables we would include in our models, we always need to assess multicollinearity, or in other words, how correlated two variables are to one another. We began with a cleaned dataset containing 32 columns and 457,670 rows. Focusing on the numeric variables, we generated a correlation matrix (see Figure 1). The results showed that most variables had weak correlations with one another, with the exception of variables measuring similar behaviors, such as sugary and non-sugary drink consumption, which were captured at different time intervals. In this case we will just need to select one manner of measuring this type of drink consumption. The only other two variables which are “moderately correlated” with one

another is Health Insurance and Medical Visit Missed Due To Cost. This is a logical correlation as populations without health insurance could need to spend substantially more money during medical appointments.

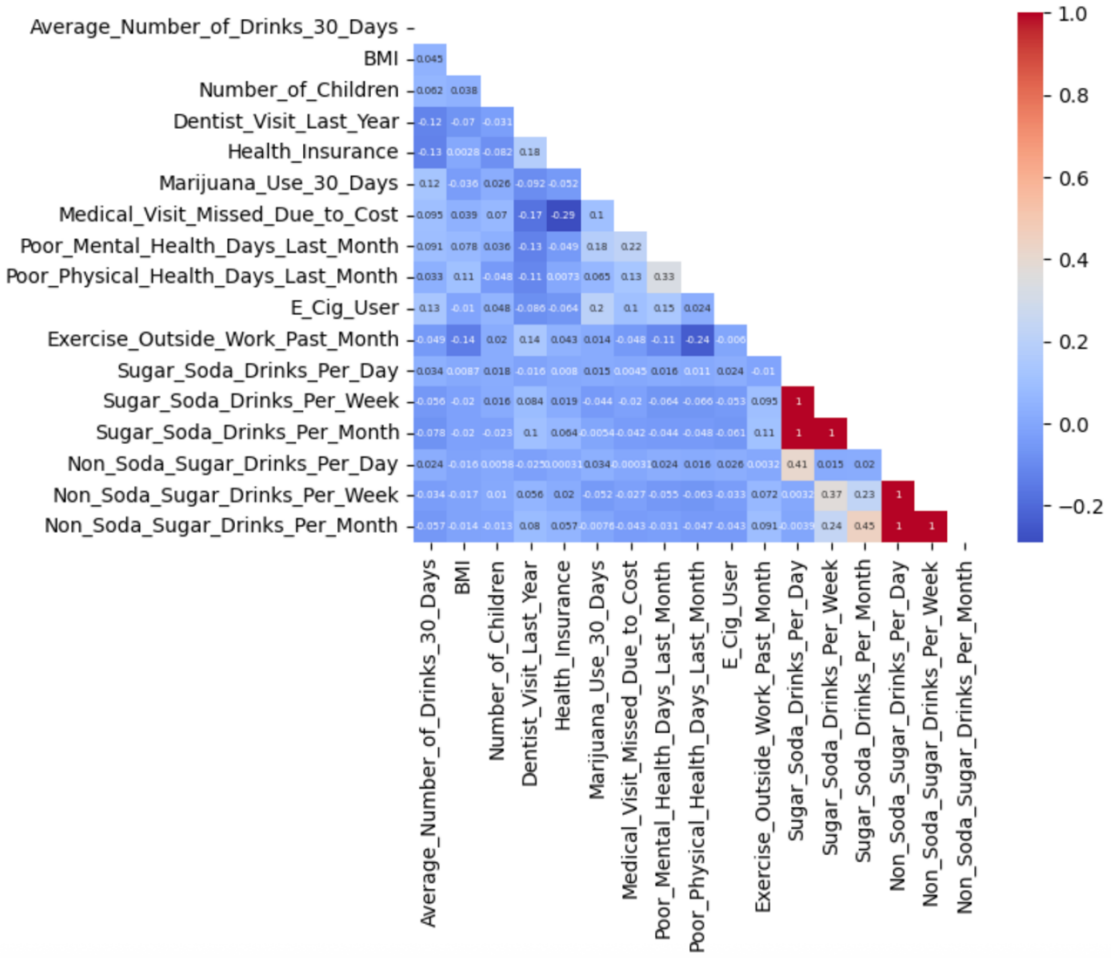

Figure 1: A correlation plot of all numeric variables, where the more red the color, the stronger the correlation

One of the more interesting findings during our Exploratory Data Analysis was that e-cigarette usage was more common in people who were Satisfied or Very Satisfied with their life, as you can see below in figure 2. While our hypothesis was that people who practiced healthier habits were less likely to consume e-cigarettes, we also had some preconceived notions that

people may use the drug as a coping mechanism, as people often do with food, alcohol and other drugs. This data point seems to point seems to suggest the opposite. It does open further questions in what other commonalities people with higher life satisfaction might have with one another and if there are cultural or societal reasons for e-cigarette usage amongst these groups?

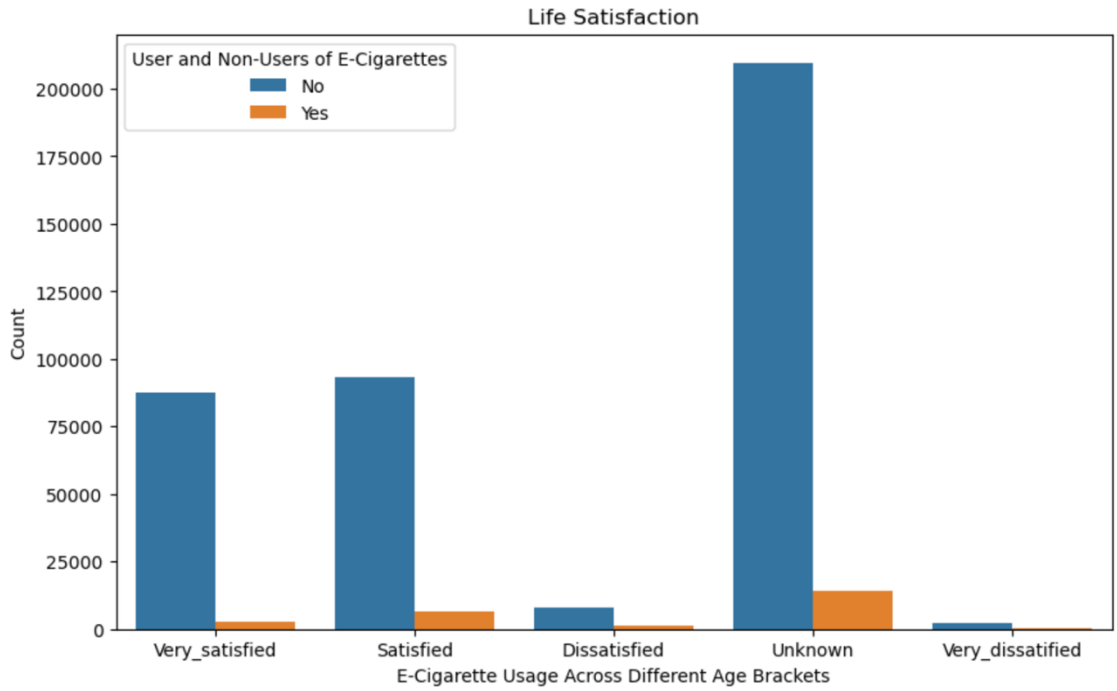

Figure 2: A look at E-cigarette usage across different buckets of self-reported life satisfaction levels

When exploring how different age brackets consume e-cigarettes, it was unsurprising to find that the younger the age group, the higher the consumption levels were. This was shown in figure 3 below. The product has only been in the US market for 20 years, so this data point aligned with our expectations. We will need to ensure that any splits of data we use to create and then test our models, include equal proportions of these age groups.

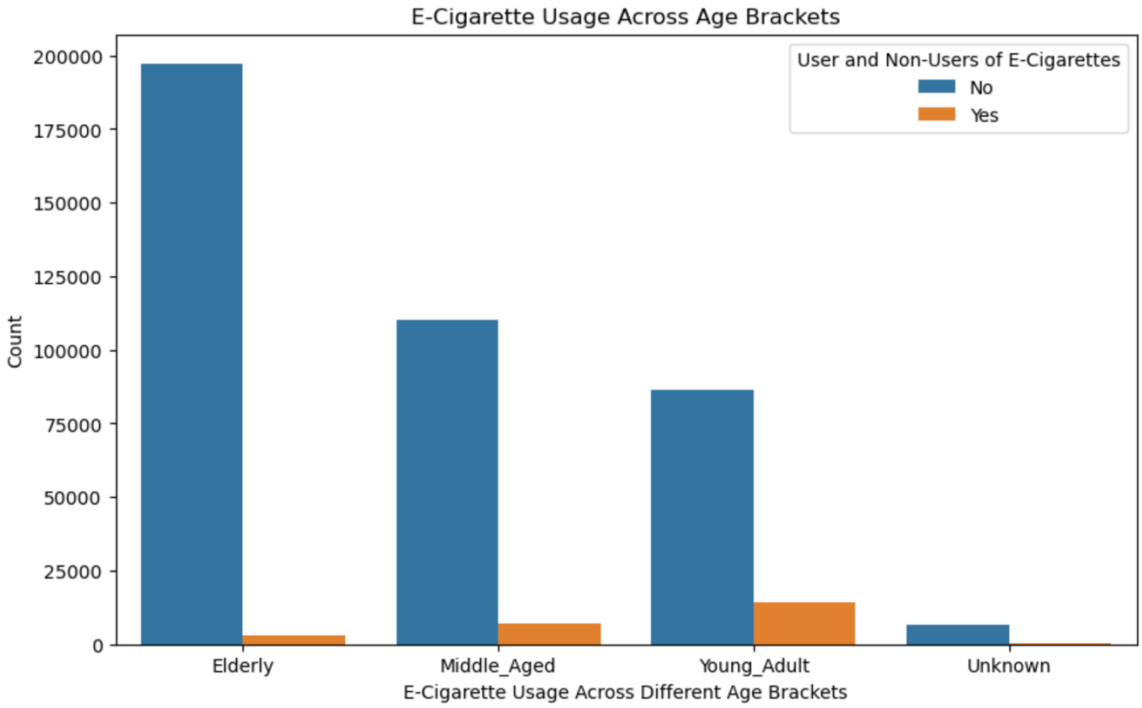

Figure 3: A look at E-cigarette usage amongst different age brackets

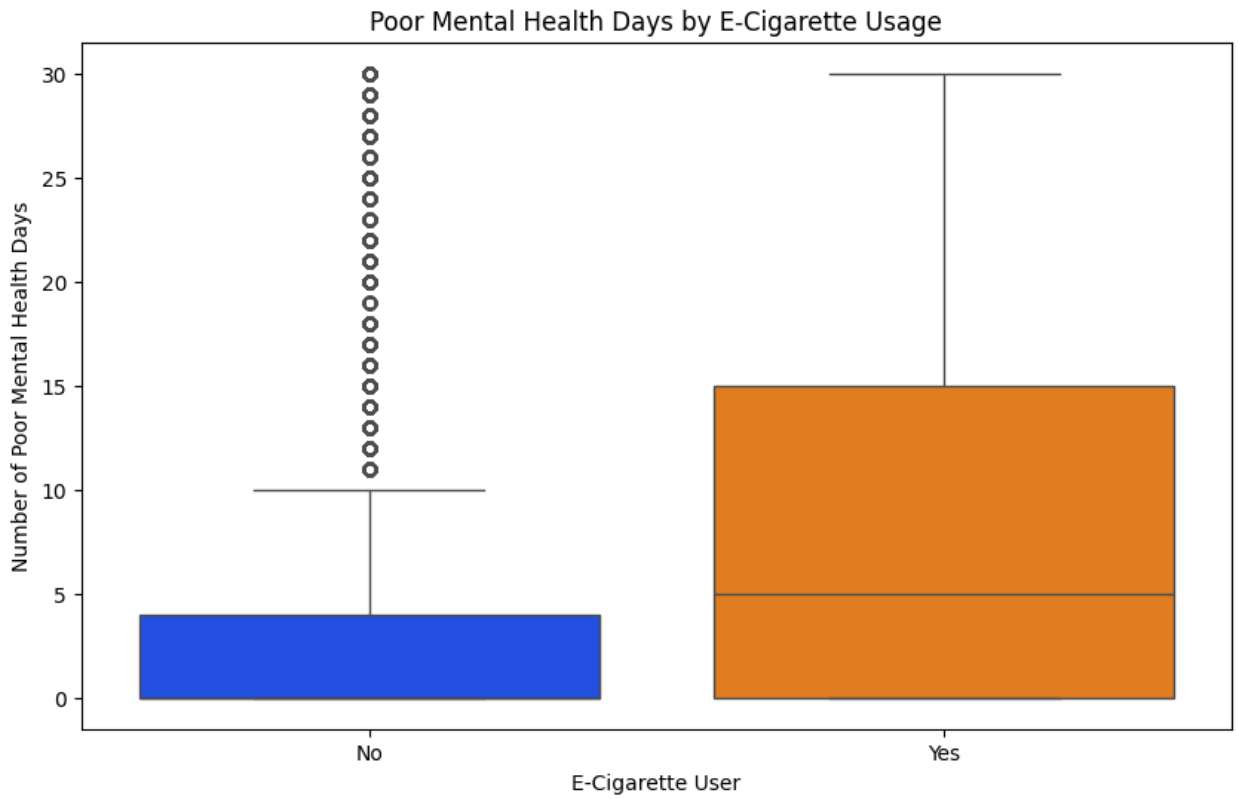

Figure 4: A Look at the number of poor mental health days amongst E-cigarette users and non-users

Figure 4 shows that e-cigarette users have a higher median number of poor mental health days than respondents who do not use e-cigarettes. There is still an overlap in the number of days, indicating that both e-cigarette users and non-e-cigarette users have poor mental health days. The data is right skewed, showing an especially large number of outliers for non-e-cigarette users. Due to the skewed data, median is a better measurement.

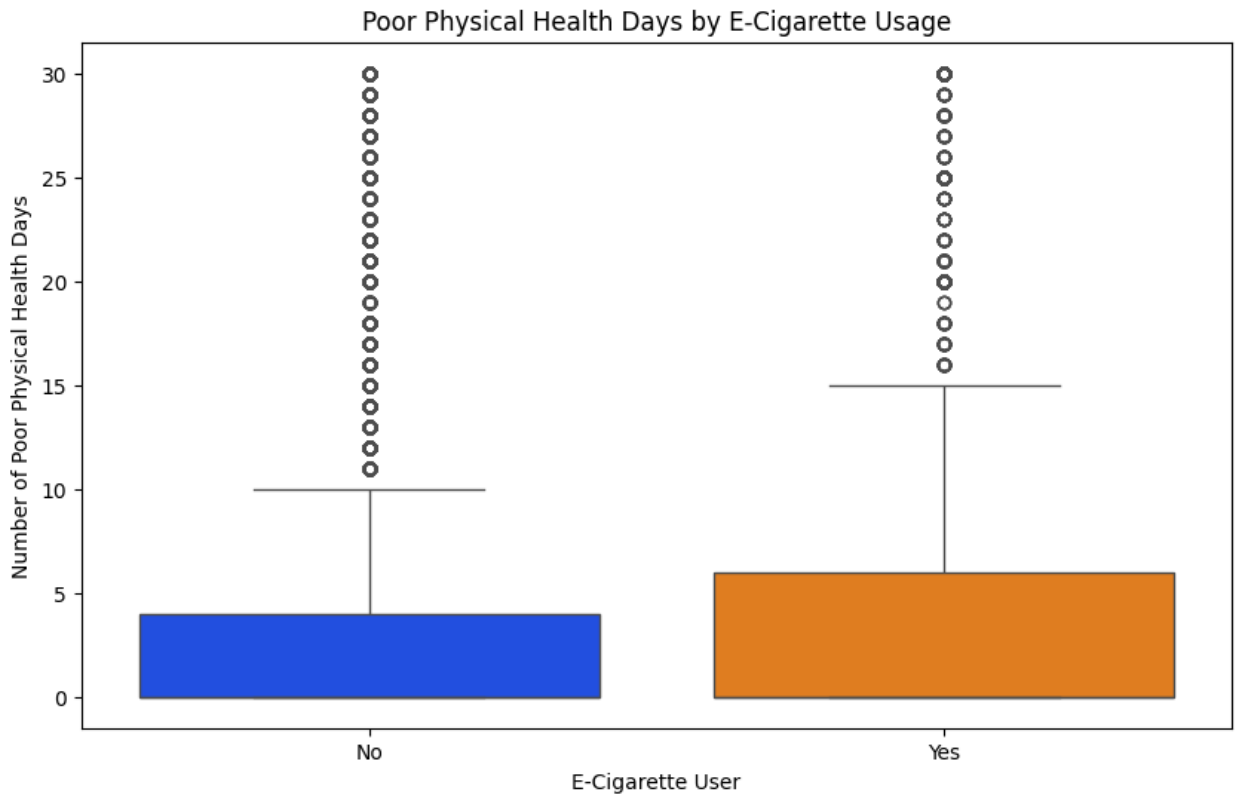

Figure 5: A Boxplot showing the number of poor physical health days by e-cigarette usage

Figure 5 shows that the number of poor physical health days among e-cigarette users and non-users is more similar than the poor mental health days. As seen in Figure 5, the data is right skewed, with a large number of outliers for both e-cigarette users and non-users. The upper quartile is higher for e-cigarette users at approximately 6 days than for non-e-cigarette users, which is approximately 4 days. Similarly, the maximum number of poor physical health days for e-cigarette users is higher, at approximately 15 days than for non-users at 10 days.

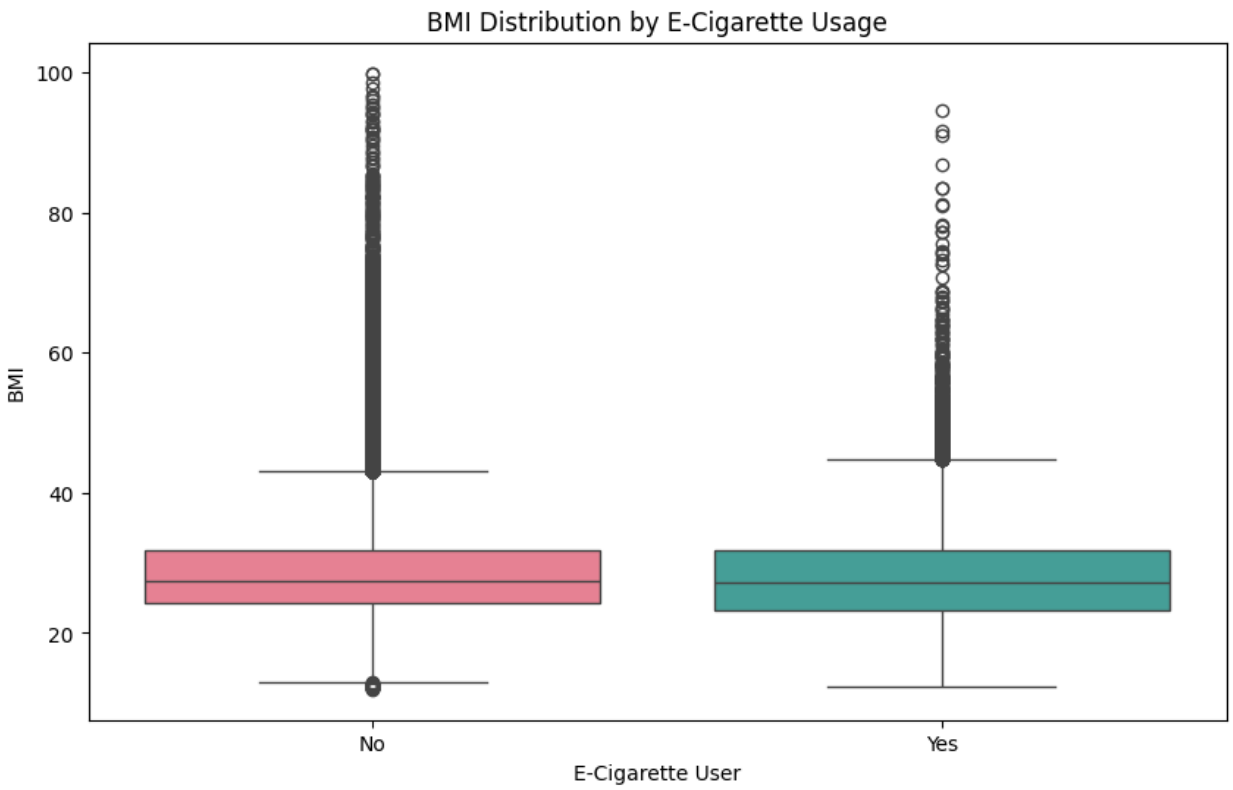

Figure 6: A boxplot showing BMI distribution by e-cigarette usage

Figure 6 shows similar BMI distributions between e-cigarette users and non-e-cigarette users. The median BMI for both groups is 27.12, with a lower quartile of approximately 23 and an upper quartile of approximately 45. There are extensive outliers for each, with some BMI values reaching 100. This could indicate extreme or incorrect records.

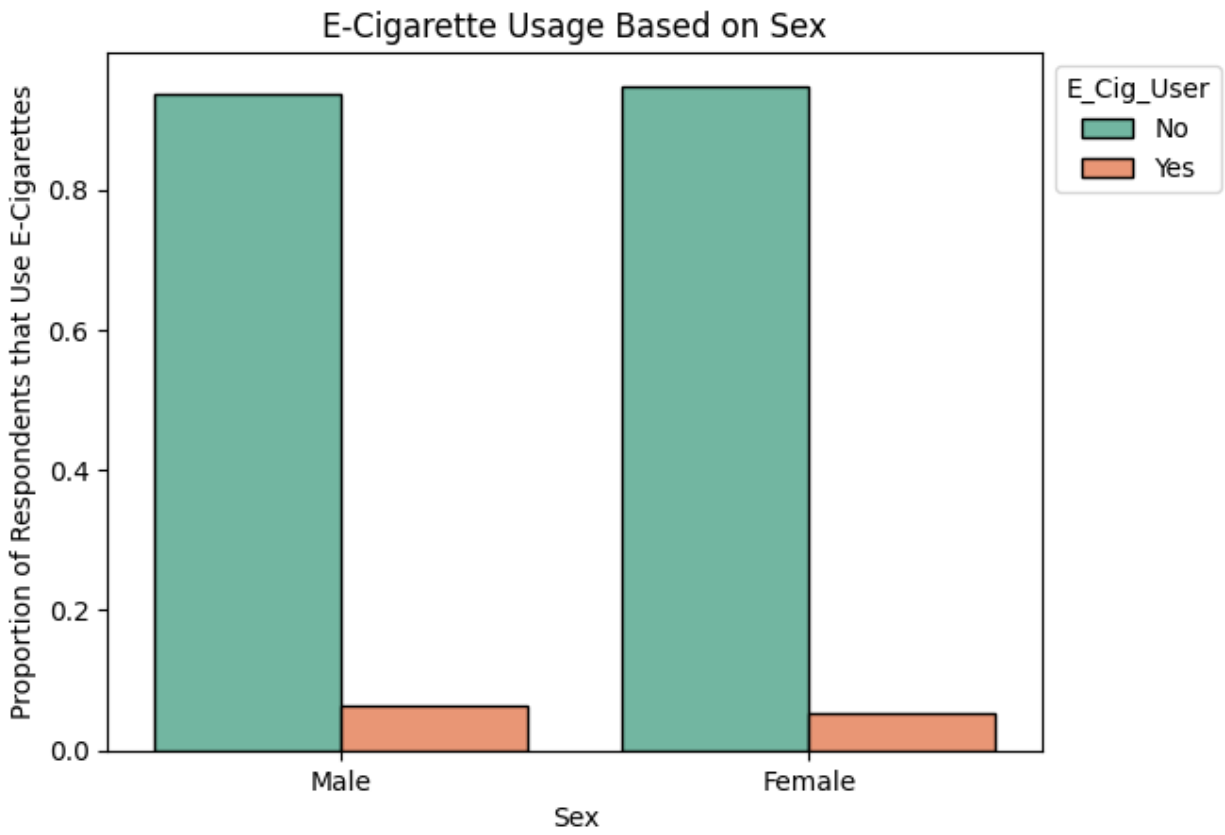

Figure 7: A barplot showing e-cigarette usage by sex

Figure 7 shows the proportions of respondents who use e-cigarettes by sex. Similarly to Figure 3, it is clear that the majority of individuals do not use them, but that males use slightly more than women.

Appendix:

categorical_missing_variables_summary

Variable	Missing Count	Missing Percent
Dentist_Visit_Last_Year	5,706	1.25
Health_Insurance	18,544	4.05
Medical_Visit_Missed_Due_to_Cost	1,673	0.37
Age_Category	0	0.0
E_Cig_User	32,751	7.16
Other_Tobacco_Usage	0	0.0
Sex	0	0.0
Race	0	0.0
Time_Since_Last_Check_Up	0	0.0
Exercise_Outside_Work_Past_Month	1,315	0.29
Loneliness	0	0.0
Health_Rating	0	0.0
Prediabetic_Diagnosis	0	0.0
Annual_Income	0	0.0
Days_Poor_Physical_Health	0	0.0
Education	0	0.0
Social_Emotional_Support	0	0.0
General_Health_Personal_Rating	0	0.0
Life_Satisfaction	0	0.0
Marital_Status	0	0.0

Table 1: Missing count and percent for categorical variables.

categorical_unknown_variables_summary

Variable	Unknown Count	Unknown Percent
Dentist_Visit_Last_Year	0	0.0
Health_Insurance	0	0.0
Medical_Visit_Missed_Due_to_Cost	0	0.0
Age_Category	8,310	1.82
E_Cig_User	0	0.0
Other_Tobacco_Usage	30,657	6.7
Sex	0	0.0
Race	9,103	1.99
Time_Since_Last_Check_Up	5,334	1.17
Exercise_Outside_Work_Past_Month	0	0.0
Loneliness	255,537	55.83
Health_Rating	1,310	0.29
Prediabetic_Diagnosis	299,317	65.4
Annual_Income	87,423	19.1
Days_Poor_Physical_Health	11,067	2.42
Education	2,363	0.52
Social_Emotional_Support	257,046	56.16
General_Health_Personal_Rating	1,310	0.29
Life_Satisfaction	255,712	55.87
Marital_Status	4,222	0.92

Table 2: Unknown count and percent for categorical variables.

continuous_variables_summary

Variable	Count	Mean	St. Dev.	Median	95% CI Lower	95% CI Upper	Min	Max	Skew	Kurtosis
Average_Number_of_Drinks_30_Days	207,713	2.26	2.42	2.0	2.25	2.27	0.0	76.0	9.31	178.85
BMI	414,633	28.56	6.59	27.44	28.54	28.58	12.0	99.84	1.42	4.74
Number_of_Children	448,291	0.48	1.14	0.0	0.48	0.49	0.0	83.0	16.75	1028.58
Marijuana_Use_30_Days	135,881	2.35	7.34	0.0	2.32	2.39	0.0	30.0	3.19	8.68
Poor_Mental_Health_Days_Last_Month	449,514	4.41	8.36	0.0	4.39	4.44	0.0	30.0	2.11	3.33
Poor_Physical_Health_Days_Last_Month	446,603	4.61	8.93	0.0	4.58	4.63	0.0	30.0	2.08	2.93
Sugar_Soda_Drinks_Per_Day	15,602	2.42	6.7	1.0	2.31	2.52	1.0	99.0	12.36	169.41
Sugar_Soda_Drinks_Per_Week	37,343	-39.6	49.35	1.0	-40.1	-39.1	-99.0	99.0	-0.34	-1.87
Sugar_Soda_Drinks_Per_Month	61,156	-83.51	80.29	-98.0	-84.15	-82.88	-199.0	99.0	-0.21	-1.34
Non_Soda_Sugar_Drinks_Per_Day	12,223	2.38	6.74	1.0	2.26	2.5	1.0	99.0	12.2	165.95
Non_Soda_Sugar_Drinks_Per_Week	30,936	-37.3	48.97	1.0	-37.85	-36.76	-99.0	99.0	-0.43	-1.8
Non_Soda_Sugar_Drinks_Per_Month	49,457	-83.89	79.33	-98.0	-84.59	-83.19	-199.0	99.0	-0.19	-1.3

Table 3: Continuous variable summary statistics

continuous_missing_variables_summary

Variable	Missing Count	Missing Percent
Average_Number_of_Drinks_30_Days	249,957	54.62
BMI	43,037	9.4
Number_of_Children	9,379	2.05
Marijuana_Use_30_Days	321,789	70.31
Poor_Mental_Health_Days_Last_Month	8,156	1.78
Poor_Physical_Health_Days_Last_Month	11,067	2.42
Sugar_Soda_Drinks_Per_Day	442,068	96.59
Sugar_Soda_Drinks_Per_Week	420,327	91.84
Sugar_Soda_Drinks_Per_Month	396,514	86.64
Non_Soda_Sugar_Drinks_Per_Day	445,447	97.33
Non_Soda_Sugar_Drinks_Per_Week	426,734	93.24
Non_Soda_Sugar_Drinks_Per_Month	408,213	89.19

Table 4: Missing count and percent for continuous variables.

GitHub Link:https://github.com/KWilson310/E-Cigarette-and-Healthy-Habits-Study

contiuous_variables_by_outcome_summary and categorical_variables_summary

Data Dictionary ->

Variable Name	Data Type	Variable Description	Variable Details	Reasoning	Keep / Cut Rationale
Age_Category	Categorical	Age category of respondents	Young_Adult Middle_Aged Elderly Unknown	Predictor variable	Keeping as a demographic variable
ALCDAY4	Numeric	Represents the number of days per week or per month the respondent had at least one drink of an alcoholic beverage	100 101-107: Days per week 201-230: Days in past 30 days 777: Don't know 888: No drinks in the past 30 days 999: Refused Blank	Predictor variable	Removed from dataset, as the column below looks better for alcohol comsumption
Average_Number_of_Drinks_30_Days	Numeric	Represents the average number of drinks consumed on a day when the respondent drank in the past 30 days	1-76: Number of drinks 0: No drinks Unknown	Predictor variable	Keeping as a lifestyle variable
BMI	Numeric	The Body Mass Index (BMI) of the respondent	1-9999: 1 or greater Blank	Predictor variable	Keeping as a physical health variable
_BMI5CAT	Categorical	BMI Categories	1: Underweight (_BMI5 < 1850) 2: Normal Weight (1850 <= _BMI5 < 2500) 3: Overweight (2500 <= _BMI5 < 3000) 4: Obese (3000 <= _BMI5 < 9999) Blank	Predictor variable	Removed from the dataset, as it was redundent. If needed, the above variable could be categorized
Time_Since_Last_Check_Up	Categorical	How long it has been since the respondent last had a routine checkup	Within_the_last_year Within_the_past_two_years Within_the_past_five_years Five_or_more_years_ago Never Unknown	Predictor variable	Keeping as a physical health variable
Number_of_Children	Numeric	Number of children under 18 years old living in the respondent's household	0 1-87: Number of children NaN	Predictor variable	Keeping as a demographic variable
E_Cig_User	Categorical	Represents whether or not the respondent is a current e-cigarette user	0: Not a current e-cigarette user 1: Is a current e-cigarette user NaN	Response/Outcome variable	
CVDCRHD4	Categorical	Whether the respondent has ever been told they have angina or coronary heart disease	1: Yes 2: No 7: Don't know 9: Refused Blank	Predictor variable	Removed from dataset
Dentist_Visit_Last_Year	Categorical	Whether the respondent has visited a dentist in the past year	1: Yes 0: No NaN	Predictor variable	Keeping as a physical health variable
DIABETE4	Categorical	Whether the respondent has ever been told they have diabetes	1: Yes 2: Yes, but only female during pregnancy 3: No 4: No (pre-diabetes or borderline) 7: Don't know 9: Refused Blank	Predictor variable	Removed from dataset
ECIGNOW3	Categorical	Represents the current e-cigarette usage status of the respondent	1: Never used e-cigarettes 2: Use e-cigarettes every day 3: Use e-cigarettes some days 4: Previous user, but not anymore 7: Don't know 9: Refused Blank	Predictor variable	Removed from dataset due to risk of data leakage
Education	Categorical	Respondent's highest level of education	Never_attended_school_or_only_kindergarten Grades_1-8_(elementary) Grades_9-11_(some high school) Grade_12_or_GED_(high school graduate) College_1-3_years_(some college or technical school) College_4_years_or_more_(college graduate) Unknown	Predictor variable	Keeping as a demographic variable
Social_Emotional_Support	Categorical	How often the respondent gets the social and emotional support that they need	Always Usually Sometimes Rarely Never Unknown	Predictor variable	Keeping as a mental health variable
Exercise_Outside_Work_Past_Month	Categorical	Whether or not the respondent has exercised outside of work in the past month	1: Yes 0: No NaN	Predictor variable	Keeping as a physical health variable
General_Health_Personal_Rating	Categorical	How the respondent thinks their general health is	Excellent Very_good Good Fair Poor Unknown	Predictor variable	Keeping as a physical health variable
Health_Insurance	Categorical	Calculated variable that represents whether or not the respondent has health insurance	1: Have some form of insurance 0: Does not have health insurance NaN	Predictor variable	Keeping as a demographic variable

Annual_Income	Categorical	Annual household income of the respondent	Less_than_$10,000 Less_than_$15,000 Less_than_$20,000 Less_than_$25,000 Less_than_$35,000 Less_than_$50,000 Less_than_$75,000 Less_than_$100,000 Less_than_$150,000 Less_than_$200,000 200,000+ Unknown	Predictor variable	Keeping as a demographic variable
LASTSMK2	Categorical	How long it has been since the respondent last smoked a cigarette	1: Within the past month 2: Within the past three months 3: Within the past six months 4: Within the past year 5: Within the past five years 6: Within the past ten years 7: Ten years or more 8: Never smoked regularly 77: Don't know 99: Refused Blank	Predictor variable	Removed from dataset due to risk of data leakage
_LCSAGE	Categorical	Calculated variable that represents lung cancer screening age groups	1: 18-49 2: 50-80 3: 81+	Predictor variable	Removed due to this only applying to a certain population
_LCSCTSN	Categorical	Whether or not the respondent has had a chest CT scan for lung cancer	1: Had a chest CT scan for lung cancer within 1 year 2: Had a chest CT scan for lung cancer over 1 year ago 3: Had a chest CT scan for other reason 4: Never had a chest CT scan Blank	Predictor variable	Removed due to this only applying to a certain population
LCSLAST_	Numeric	Calculated variable representing the age the respondent was when they last smoked cigarettes regularly	1-80: Age in years 81: Age in years (81+) Blank	Predictor variable	Removed from dataset due to risk of data leakage
LCSNUMC_	Numeric	When smoking regularly, how many cigarettes the respondent smokes per day	1-300: Number of cigarettes Blank	Predictor variable	Removed from dataset due to risk of data leakage
_LCSYSMK	Numeric	The number of years the respondent smoked cigarettes	0-100: Number of years Blank	Predictor variable	Removed from dataset due to risk of data leakage
_LCSSMKG	Categorical	LCS (Lung Cancer Screening) Smoking Group status	1: Current smoker (20+ packs per year) 2: Former smoker (20+ packs per year), quit < 15 years 3: Current smoker (less than 20 packs per year) 4: Former smoker (20+ packs per year), quit >= 15 years 5: Former smoker (< 20 packs per year) 6: Never smoked Blank	Predictor variable	Removed from dataset due to risk of data leakage
_LCSYQTS	Numeric	Number of years since the respondent quit smoking	0-100: Number of years since they last smoked cigarettes Blank Hidden	Predictor variable	Removed from dataset due to risk of data leakage
Life_Satisfaction	Categorical	Level of satisfaction in the respondent's life	Very_satisfied Satisfied Dissatisfied Very dissatisfied Unknown	Predictor variable	Keeping as a mental health variable
Marijuana_Use_30_Days	Numeric	How many days the respondent has used marijuana or cannabis in the past 30 days	0-30: Number of Days NaN	Predictor variable	Keeping as a lifestyle variable
Marital_Status	Categorical	Marital status of the respondent	Married Divorced Widowed Separated Never_Married Member_of_unmarried_couple Unknown	Predictor variable	Keeping as a demographic variable
Medical_Visit_Missed_Due_to_Cost	Categorical	Whether or not there was a time in the past 12 months the respondent needed to see a doctor but couldn't because of the cost	1: Yes 0: No NaN	Predictor variable	Keeping as a demographic variable
MENTCIGS	Categorical	Whether the respondent usually uses menthol cigarettes when currently smoking	1: Yes 2: No 7: Don't know 9: Refused Blank	Predictor variable	Removed from dataset due to risk of data leakage
MENTECIG	Categorical	Whether the respondent usually uses menthol e-cigarettes when currently using e-cigarettes	1: Yes 2: No 7: Don't know 9: Refused Blank	Predictor variable	Removed from dataset due to risk of data leakage
_MENT14D	Categorical	Number of days the respondent had poor mental health	1: 0 days 2: 1-13 days 3: 14+ days 9: don't know/refused/missing	Predictor variable	Removed from dataset. Will investigate further to determine if it should be added back in
Poor_Mental_Health_Days_Last_Month	Numeric	Indicates the number of days in the past 30 days the respondent's mental health was not good	1-30: Number of Days 0: None NaN	Predictor variable	Keeping as a mental health variable

_PACKDAY	Numeric	Calculated variable that represents the number of packs of cigarettes the respondent smokes per day	0-100: Number of packs smoked per day Blank	Predictor variable	Removed from dataset due to risk of data leakage
_PACKYRS	Numeric	Calculated variable that represents the number of years the individual has smoked their reported number of packs per day	0-999: Number of years Blank	Predictor variable	Removed from dataset due to risk of data leakage
Days_Poor_Physical_Health	Categorical	Number of days the respondent had poor physical health	0_days 1-13_days 14+_days Unknown	Predictor variable	Keeping as a physical health variable
Poor_Physical_Health_Days_Last_Month	Numeric	Indicates the number of days in the past 30 days the respondent's physical health was not good	1-30: Number of Days 0: None NaN	Predictor variable	Keeping as a physical health variable
POORHLTH	Numeric	Indicates number of days in the past 30 days the respondent had poor physical or mental health that kept them from doing their regular activities	1-30: Number of Days 88: None 77: Don't know 99: Refused Blank	Predictor variable	Removed due to redundency
Prediabetic_Diagnosis	Categorical	Whether the respondent has ever been told they have pre-diabetes	Yes Yes_during_Pregnancy No Unknown	Predictor variable	Keeping as a physical health variable
Race	Categorical	Race of the respondent	White Black American Indian or Alaskan Native Asian Native Hawaiian or other Pacific Islander Other Race Multiracial Hispanic Unknown	Predictor variable	Keeping as a demographic variable
_RFBMI5	Categorical	Respondents with a BMI > 25, classifying them as overweight or obese	1: No 2: Yes 9: Blank	Predictor variable	Removed due to redundency
Health_Rating	Categorical	A calculated variable representing what respondents think of their health	Good_or_Better_Health Fair_or_Poor_Health Unknown	Predictor variable	Keeping as a physical health variable
_RFSMOK3	Categorical	Current smoking status of respondents	1: No 2: Yes 9: Don't know	Predictor variable	Removed from dataset due to risk of data leakage
Sex	Categorical	Sex of the respondent	Male Female	Predictor variable	Keeping as a demographic variable
Loneliness	Categorical	How often the respondent feels lonely	Always Usually Sometimes Rarely Never Unknown	Predictor variable	Keeping as a mental health variable
SMOKE100	Categorical	Whether or not the respondent has smoked at least 100 cigarettes in their lifetime	1: Yes 2: No 7: Don't Know 9: Refused Blank	Predictor variable	Removed from dataset due to risk of data leakage
SMOKDAY2	Categorical	Whether or not the respondent currently smokes cigarettes every day, some days, or not at all	1: Every day 2: Some days 3: Not at all 7: Don't Know 9: Refused Blank	Predictor variable	Removed from dataset due to risk of data leakage
_SMOKER3	Categorical	Smoker status	1: Current smoker (smokes every day) 2: Current smoker (smokes some days) 3: Former smoker 4: Never smoked 9: Don't know Blank	Predictor variable	Removed from dataset due to risk of data leakage
Sugar_Soda_Drinks_Per_Day	Numeric	Derived from SSBSUGR. Represents the number of soda drinks the respondent has per day	1-99 NaN	Predictor variable	Created as a lifestyle variable
Sugar_Soda_Drinks_Per_Week	Numeric	Derived from SSBSUGR. Represents the number of soda drinks the respondent has per week	-99-99 NaN	Predictor variable	Created as a lifestyle variable. Need to investigate further as there are negative values
Sugar_Soda_Drinks_Per_Month	Numeric	Derived from SSBSUGR. Represents the number of soda drinks the respondent has per month	-199-99 NaN	Predictor variable	Created as a lifestyle variable. Need to investigate further as there are negative values
SSBSUGR2	Numeric	The number of times the respondent drank soda or pop that contained sugar in the past 30 days	101-199: Times per day 201-299: Times per week 301-399: Times per month 777: Don't know 888: Never 999: Refused Blank	Predictor variable	Created three new variables from this and removed the original
Non_Soda_Sugar_Drinks_Per_Day	Numeric	Derived from SSBFRUT3. Represents the number of non-soda sugar drinks the respondent has per day	1-99 NaN	Predictor variable	Created as a lifestyle variable

Non_Soda_Sugar_Drinks_Per_Week	Numeric	Derived from SSBFRUT3. Represents the number of non-soda sugar drinks the respondent has per week	-99-99 NaN	Predictor variable	Created as a lifestyle variable. Need to investigate further as there are negative values
Non_Soda_Sugar_Drinks_Per_Month	Numeric	Derived from SSBFRUT3. Represents the number of non-soda sugar drinks the respondent has per month	-199-99 NaN	Predictor variable	Created as a lifestyle variable. Need to investigate further as there are negative values
SSBFRUT3	Numeric	The number of times the respondent drank fruit juice, sweet tea, sports, or energy drinks that contained sugar in the past 30 days	101-199: Times per day 201-299: Times per week 301-399: Times per month 777: Don't know 888: Never 999: Refused Blank	Predictor variable	Created three new variables from this and removed the original
STOPSMK2	Categorical	Whether the respondent has stopped smoking for one day or longer when trying to quit in the past 12 months	1: Yes 2: No 7: Don't know 9: Refused Blank	Predictor variable	Removed from dataset due to risk of data leakage
_TOTINDA	Categorical	A calculated variable that represents whether or not the respondent has performed physical activity or exercised in the past 30 days outside of their job	1: Had physical activity or exercise (EXERANY2 = 1) 2: No physical activity or exercise in last 30 days (EXERANY2 = 2) 9: Don't know/Refused/Missing (EXERANY2 = 7 or 9 or Missing)	Predictor variable	Removed due to redundency
Other_Tobacco_Usage	Categorical	How often the respondent currently uses chewing tobacco, snuff, or snus	Every Day Some days Not at all Unknown	Predictor variable	Keeping as a lifestyle variable

continuous_variables_by_outcome_summary

Variable	Outcome Class	Count	Mean	St. Dev.	Median	95% CI Lower	95% CI Upper	Min	Max	Skew	Kurtosis
Average_Number_of_Drinks_30_Days	0.0	192,599	2.17	2.25	2.0	2.16	2.18	0.0	76.0	9.12	173.88
Average_Number_of_Drinks_30_Days	1.0	14,504	3.44	3.8	3.0	3.38	3.5	0.0	76.0	8.37	123.86
Average_Number_of_Drinks_30_Days		610	2.64	2.77	2.0	2.42	2.86	0.0	30.0	4.46	29.55
BMI	0.0	375,218	28.59	6.54	27.44	28.57	28.61	12.0	99.84	1.41	4.67
BMI	1.0	23,558	28.31	7.16	27.12	28.22	28.4	12.4	94.61	1.39	4.13
BMI		15,857	28.1	6.64	27.12	28.0	28.21	12.05	94.94	1.73	7.35
Number_of_Children	0.0	397,669	0.47	1.11	0.0	0.46	0.47	0.0	83.0	16.79	1056.8
Number_of_Children	1.0	24,380	0.7	1.17	0.0	0.68	0.71	0.0	22.0	2.83	20.07
Number_of_Children		26,242	0.55	1.44	0.0	0.53	0.57	0.0	82.0	22.78	1170.47
Marijuana_Use_30_Days	0.0	128,346	2.0	6.79	0.0	1.97	2.04	0.0	30.0	3.54	11.12
Marijuana_Use_30_Days	1.0	7,038	8.75	12.41	0.0	8.46	9.04	0.0	30.0	0.94	-0.93
Marijuana_Use_30_Days		497	2.7	7.64	0.0	2.03	3.37	0.0	30.0	2.95	7.32
Poor_Mental_Health_Days_Last_Month	0.0	393,732	4.13	8.08	0.0	4.1	4.15	0.0	30.0	2.23	3.93
Poor_Mental_Health_Days_Last_Month	1.0	24,050	9.39	10.9	5.0	9.25	9.53	0.0	30.0	0.9	-0.68
Poor_Mental_Health_Days_Last_Month		31,732	4.19	8.36	0.0	4.1	4.28	0.0	30.0	2.18	3.59
Poor_Physical_Health_Days_Last_Month	0.0	391,141	4.56	8.9	0.0	4.53	4.59	0.0	30.0	2.09	3.01
Poor_Physical_Health_Days_Last_Month	1.0	24,043	5.47	9.3	0.0	5.35	5.59	0.0	30.0	1.8	1.87
Poor_Physical_Health_Days_Last_Month		31,419	4.52	8.95	0.0	4.42	4.61	0.0	30.0	2.09	2.96
Sugar_Soda_Drinks_Per_Day	0.0	13,945	2.37	6.65	1.0	2.26	2.48	1.0	99.0	12.57	174.15
Sugar_Soda_Drinks_Per_Day	1.0	1,583	2.9	7.28	2.0	2.54	3.26	1.0	99.0	10.71	132.18
Sugar_Soda_Drinks_Per_Day		74	1.81	1.29	1.0	1.52	2.1	1.0	7.0	2.06	4.29
Sugar_Soda_Drinks_Per_Week	0.0	34,068	-38.79	49.21	1.0	-39.31	-38.27	-99.0	99.0	-0.37	-1.84
Sugar_Soda_Drinks_Per_Week	1.0	3,110	-48.23	50.06	-82.0	-49.99	-46.47	-99.0	30.0	0.02	-1.99
Sugar_Soda_Drinks_Per_Week		165	-42.95	49.98	1.0	-50.58	-35.32	-99.0	9.0	-0.21	-1.98
Sugar_Soda_Drinks_Per_Month	0.0	56,441	-82.07	79.91	-98.0	-82.73	-81.42	-199.0	99.0	-0.24	-1.32
Sugar_Soda_Drinks_Per_Month	1.0	4,475	-100.96	82.89	-98.0	-103.39	-98.53	-199.0	99.0	0.13	-1.41
Sugar_Soda_Drinks_Per_Month		240	-96.68	80.41	-99.0	-106.85	-86.5	-199.0	30.0	0.03	-1.37
Non_Soda_Sugar_Drinks_Per_Day	0.0	10,894	2.31	6.57	1.0	2.19	2.44	1.0	99.0	12.6	176.5
Non_Soda_Sugar_Drinks_Per_Day	1.0	1,265	2.89	7.69	2.0	2.46	3.31	1.0	99.0	9.99	115.8
Non_Soda_Sugar_Drinks_Per_Day		64	3.27	12.23	1.0	0.27	6.26	1.0	99.0	7.86	62.44
Non_Soda_Sugar_Drinks_Per_Week	0.0	27,981	-36.77	48.84	1.0	-37.35	-36.2	-99.0	99.0	-0.46	-1.78
Non_Soda_Sugar_Drinks_Per_Week	1.0	2,803	-42.44	49.91	1.0	-44.28	-40.59	-99.0	60.0	-0.21	-1.94
Non_Soda_Sugar_Drinks_Per_Week		152	-39.59	49.54	1.0	-47.46	-31.71	-99.0	7.0	-0.35	-1.9
Non_Soda_Sugar_Drinks_Per_Month	0.0	45,160	-82.79	79.07	-98.0	-83.52	-82.06	-199.0	99.0	-0.21	-1.28
Non_Soda_Sugar_Drinks_Per_Month	1.0	4,091	-95.17	81.26	-98.0	-97.66	-92.68	-199.0	99.0	0.04	-1.34
Non_Soda_Sugar_Drinks_Per_Month		206	-100.67	78.46	-99.0	-111.38	-89.95	-199.0	99.0	0.18	-1.08

categorical_variables_summary

Variable	Value	Count	Percent
Dentist_Visit_Last_Year	1.0	311,259	68.01
Dentist_Visit_Last_Year	0.0	140,705	30.74
Dentist_Visit_Last_Year		5,706	1.25
Health_Insurance	1.0	413,720	90.4
Health_Insurance	0.0	25,406	5.55
Health_Insurance		18,544	4.05
Medical_Visit_Missed_Due_to_Cost	0.0	412,634	90.16
Medical_Visit_Missed_Due_to_Cost	1.0	43,363	9.47
Medical_Visit_Missed_Due_to_Cost		1,673	0.37
Age_Category	Elderly	214,421	46.85
Age_Category	Middle_Aged	126,501	27.64
Age_Category	Young_Adult	108,438	23.69
Age_Category	Unknown	8,310	1.82
E_Cig_User	0.0	400,413	87.49
E_Cig_User		32,751	7.16
E_Cig_User	1.0	24,506	5.35
Other_Tobacco_Usage	Not at all	414,356	90.54
Other_Tobacco_Usage	Unknown	30,657	6.7
Other_Tobacco_Usage	Every Day	7,367	1.61
Other_Tobacco_Usage	Some days	5,290	1.16
Sex	Female	240,183	52.48
Sex	Male	217,487	47.52
Race	White	329,346	71.96
Race	Hispanic	48,646	10.63
Race	Black	35,172	7.69
Race	Asian	12,646	2.76
Race	Multiracial	10,486	2.29
Race	Unknown	9,103	1.99
Race	American Indian or Alaskan Native	6,460	1.41

Race	Other Race	3,737	0.82
Race	Native American or other Pacific Islander	2,074	0.45
Time_Since_Last_Check_Up	Within_the_last_year	370,023	80.85
Time_Since_Last_Check_Up	Within_the_past_two_years	39,655	8.66
Time_Since_Last_Check_Up	Within_the_past_five_years	21,480	4.69
Time_Since_Last_Check_Up	Five_or_more_years_ago	18,395	4.02
Time_Since_Last_Check_Up	Unknown	5,334	1.17
Time_Since_Last_Check_Up	Never	2,783	0.61
Exercise_Outside_Work_Past_Month	1.0	350,061	76.49
Exercise_Outside_Work_Past_Month	0.0	106,294	23.23
Exercise_Outside_Work_Past_Month		1,315	0.29
Loneliness	Unknown	255,537	55.83
Loneliness	Never	76,943	16.81
Loneliness	Rarely	65,222	14.25
Loneliness	Sometimes	47,449	10.37
Loneliness	Usually	7,455	1.63
Loneliness	Always	5,064	1.11
Health_Rating	Good_or_Better_Health	366,237	80.02
Health_Rating	Fair_or_Poor_Health	90,123	19.69
Health_Rating	Unknown	1,310	0.29
Prediabetic_Diagnosis	Unknown	299,317	65.4
Prediabetic_Diagnosis	No	131,009	28.63
Prediabetic_Diagnosis	Yes	25,796	5.64
Prediabetic_Diagnosis	Yes_during_Pregnancy	1,548	0.34
Annual_Income	Unknown	87,423	19.1
Annual_Income	Less_than_$75,000	60,709	13.26
Annual_Income	Less_than_$150,000	55,778	12.19
Annual_Income	Less_than_$100,000	51,700	11.3
Annual_Income	Less_than_$50,000	49,686	10.86
Annual_Income	Less_than_$35,000	41,702	9.11
Annual_Income	$200,000.00	29,694	6.49
Annual_Income	Less_than_$200,000	26,675	5.83

Annual_Income	Less_than_$25,000	19,841	4.34
Annual_Income	Less_than_$20,000	13,662	2.99
Annual_Income	Less_than_$15,000	10,542	2.3
Annual_Income	Less_than_$10,000	10,258	2.24
Days_Poor_Physical_Health	0_Days	267,576	58.46
Days_Poor_Physical_Health	1-13_Days	115,200	25.17
Days_Poor_Physical_Health	14+_Days	63,827	13.95
Days_Poor_Physical_Health	Unknown	11,067	2.42
Education	College_4_years_or_more_(college_gradu	191,549	41.85
Education	College_1-3_years_(some_college_or_tecl	120,732	26.38
Education	Grade_12_or_GED_(high_school_graduate	115,877	25.32
Education	Grades_9-11_(some_high_school)	17,200	3.76
Education	Grades_1-8_(elementary)	9,226	2.02
Education	Unknown	2,363	0.52
Education	Never_attended_school_or_only_Kinderga	723	0.16
Social_Emotional_Support	Unknown	257,046	56.16
Social_Emotional_Support	Always	96,541	21.09
Social_Emotional_Support	Usually	59,933	13.1
Social_Emotional_Support	Sometimes	27,111	5.92
Social_Emotional_Support	Rarely	8,670	1.89
Social_Emotional_Support	Never	8,369	1.83
General_Health_Personal_Rating	Good	156,229	34.14
General_Health_Personal_Rating	Very_good	145,791	31.86
General_Health_Personal_Rating	Fair	67,919	14.84
General_Health_Personal_Rating	Excellent	64,217	14.03
General_Health_Personal_Rating	Poor	22,204	4.85
General_Health_Personal_Rating	Unknown	1,310	0.29
Life_Satisfaction	Unknown	255,712	55.87
Life_Satisfaction	Satisfied	100,151	21.88
Life_Satisfaction	Very_satisfied	90,530	19.78
Life_Satisfaction	Dissatisfied	8,832	1.93
Life_Satisfaction	Very_dissatified	2,445	0.53

Marital_Status	Married	229,393	50.12
Marital_Status	Never_Married	84,851	18.54
Marital_Status	Divorced	58,827	12.85
Marital_Status	Widowed	49,161	10.74
Marital_Status	Member_of_unmarried_couple	21,754	4.75
Marital_Status	Separated	9,462	2.07
Marital_Status	Unknown	4,222	0.92